Project Update: Multi-Agent Housing and Insurance Market

Neil Dhawan, Jon Legasa, Alejandro Ordonez

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Abstract

We have constructed a housing and insurance environment for Q-learning agents using Lorenz allocations of income and ownership, spatially correlated home prices, and income-dependent residence assignment. Poisson disasters destroy homes within fixed zones, linking property losses and displacement to insurance claims. The proposed spatial extension assigns hidden regional risk that agents learn from public disaster histories. Three experimental worlds—"Peace", "Standard", and "Chaos"—compare equal and unequal initial resources under constant or increasing disaster rates.

1 Market model

Agents act as renters, owner-occupiers, or landlords; insurers offer coverage. Their decisions connect ownership, rent, insurance purchases, and claims.

Lorenz curves control income and ownership inequality. A shared ranking associates higher income with larger intended portfolios. Spatial price variation and noisy residence assignment create neighborhoods with different wealth and exposure.

Disasters produce losses across neighboring homes, exposing owners, residents, and insurers to the same event.

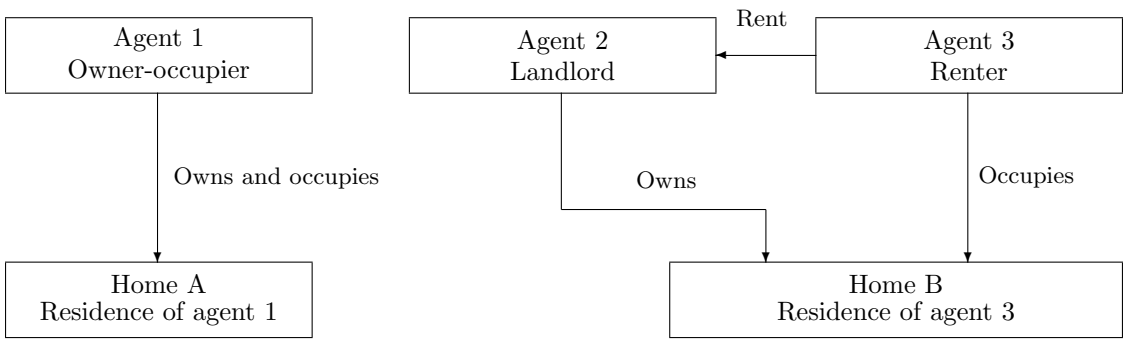


Figure 1: Housing relationships: agent 1 owns their residence; agent 3 rents from agent 2. Roles can overlap or change.

2 Model assumptions

- **Agents and objectives.** Individual and insurer agents use Q-learning to maximize expected discounted cumulative rewards. Renter, owner-occupier, and landlord are housing relationships, not fixed agent types.
- **Housing and allocation.** Each grid cell is a region, not a home. Every region holds at least one home, larger regions hold more homes, and region sizes follow a smoothed spatial field. Homes in a region share its location, and their initial prices vary around the regional mean, and they have no separate positions inside it. Agents are tied to homes rather than to cells. The population and housing supply are fixed and equal, with one resident per home initially. Ownership is separate from residence and may be concentrated. Initial allocation does not enforce affordability. Initial prices are positive; rent and starting capital are configured separately.
- **Income and inequality.** External income is assigned once, accrues each period, and is paid every 10 completed simulation runs; landlords receive accrued rent every 20 completed runs. Section 6.1 defines the payment clock. Jobs are not simulated. Income and ownership curves share a agent ranking. Recurring income adds money to the economy. "Peace" imposes equal initial resources, with no ongoing wealth cap or redistribution.
- **Spatial fields.** Seeded noise generates initial prices, region sizes, and a disaster-risk field. These three fields are independent of one another and remain fixed during an episode. Spatial boundaries do not wrap. A global seed supplies separate random streams for reproducible runs.
- **Disaster occurrence.** Each period, draw the number of disasters from a Poisson distribution. Choose each disaster's center independently using the hidden risk map. Draw its size independently from a Tweedie distribution, so most disasters strike a single region and large ones are rare outliers. Each event affects a

square of regions, clipped at the grid edge. The event rate may change over time; the risk map and size distribution stay fixed.

- **Public information.** After each disaster, every agent and insurer observes its time, affected cells, and resulting housing damage across the grid. The underlying risk field and location probabilities are not provided to agents. They can learn about regional risk from the shared event history, including events outside their own holdings.
- **Damage and recovery.** Every intact home in an affected region loses all modeled value and becomes uninhabitable. Its resident is displaced and its owner retains title. Surviving land value, rebuilding, and replacement housing are omitted. Repeated damage to an already destroyed home generates no new building loss.
- **Insurance and timing.** Coverage is chosen before exposure, with up-front premiums and fixed contract terms. Income and rent accrue during each run and settle on their scheduled run boundaries; when both are due, income is paid before rent. Disasters and claims continue within each run. Claims use pre-loss building values or specified contents amounts, subject to deductibles, coverage limits, and available insurer capital. Capital shortfalls are recorded; claims are settled after each event.

3 Environment and notation

The $m \times n$ grid contains mn regions, H homes, $N = H$ agents, and F insurers. Region (a, b) holds $H_{ab} \geq 1$ homes, and ρ is the mean number of homes per region. Matrices index regions by location; vectors index homes or agents.

3.1 Definitions and glossary

Let i index agents, ω homes, t simulation periods, and (a, b) and (u, v) region locations. Home ω lies in region (a_ω, b_ω). Their domains are

$$\begin{aligned} &m, n, N, F, \rho \in \mathbb{Z}_{>0}, && H = \rho mn = \sum_{a=1}^{m} \sum_{b=1}^{n} H_{ab}, \\ &i \in \{1, \ldots, N\}, && \omega \in \{1, \ldots, H\}, \qquad t \in \mathbb{Z}_{\geq 0}, \\ &(a, b), (u, v) \in \{1, \ldots, m\} \times \{1, \ldots, n\}. \end{aligned} \tag{1}$$

Quantities are current unless stated otherwise.

Symbol	Meaning
P	$\mathbb{R}_{\geq 0}^{H}$. Home values.
O	$\{1, \ldots, N\}^{H}$. Owners, represented by agent indices.
R	$\{0, \ldots, N\}^{H}$. Residents; zero denotes vacancy.
D	$\{0, 1\}^{H}$. Destruction indicators; one denotes an uninhabitable home.
$\mathbf{y}$	$\mathbb{R}_{\geq 0}^{N}$. Fixed annual external incomes.
$\mathbf{c}$	$\mathbb{R}_{\geq 0}^{N}$. Cash balances.

Initially, $P \in \mathbb{R}_{>0}^{H}$, $D = 0$, and each agent appears once in R. Ownership indices may repeat. A tenancy occurs where $R_\omega > 0$ and $R_\omega \neq O_\omega$. After displacement, a agent may have no residence.

The ownership indicator $\mathbf{1}_{\{O_\omega = i\}}$ identifies homes belonging to agent i. Their home count and property wealth are

$$\begin{aligned} \text{home count of agent } i &= \sum_{\omega=1}^{H} \mathbf{1}_{\{O_\omega = i\}}, \\ \text{property wealth of agent } i &= \sum_{\omega=1}^{H} P_\omega \mathbf{1}_{\{O_\omega = i\}}. \end{aligned} \tag{2}$$

$$\underbrace{\begin{bmatrix} 200 & 300 & 400 & 500 \end{bmatrix}}_{P \text{ (thousands)}} \qquad \underbrace{\begin{bmatrix} 3 & 4 & 3 & 4 \end{bmatrix}}_{O \text{ (owners)}} \qquad \underbrace{\begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix}}_{R \text{ (residents)}}$$

Figure 2: Illustrative allocation of four homes, listed by ω: agents 1 and 2 rent from 3 and 4. Each owner holds two homes, worth 600 and 800 thousand respectively.

Contracts link agents, insurers, homes, and coverage terms.

3.2 Learning terminology and notation

The proposed tabular Q-learning specification in Section 10 uses *household* for an individual agent and *firm* for an insurer. These are the two learner types; housing roles describe a household's relationship to homes. Reinforcement learning (RL) updates action values from rewards observed through interaction with the environment. A tabular Q-learner stores these values in a table indexed by state and action.

Symbol or term	Meaning
k	$\{h, f\}$. Learner type: household (h) or insurer (f); these superscripts label types.
j	$\{1, \ldots, F\}$. Insurer index in the learning specification.
s, s'	Type-specific states. Current and next state; s_i^h and s_j^f denote household and insurer states.
$\mathcal{A}^k$	Type-specific action set. Actions for learner type k; a is the chosen action and a' indexes candidate next actions. Here a denotes an action, rather than a grid coordinate.
$Q^k(s, a)$	$\mathbb{R}$. Estimated discounted return for a state–action pair, stored in the shared table for type k.
α	$(0, 1]$. Learning rate controlling the size of a Q-value update.
γ	$[0, 1]$. Discount factor weighting future rewards.
ε	$[0, 1]$. Exploration probability in ε-greedy action selection; otherwise an action with maximal Q-value is selected.
r, r_i^h, r_j^f	$\mathbb{R}$. Immediate reward, with household and insurer forms defined in the learning specification.
Δ	Change operator. Change in a quantity over a period, as used in the household reward.
κ, κ_f	$\mathbb{R}_{\geq 0}$. Household displacement and insurer insolvency penalties.
r_f	$\mathbb{R}_{\geq 0}$. Risk-free rate applied to insurer capital per period; distinct from the insurer reward r_j^f.
Pure premium	Expected loss cost. Fitted loss cost used as the reference for the insurer's premium ladder.
Tweedie GLM	Statistical model. Tweedie generalized linear model used in the proposed offline pure-premium fit.

3.3 Distribution functions and parameters

Symbol	Definition
L_y, L_h	$[0, 1] \to [0, 1]$. Lorenz functions for income and home counts along the same agent ordering.
μ	$\mathbb{R}_{>0}$. Mean annual external income.
Δt	$\mathbb{R}_{>0}$. Period length in years.
q	$\mathbb{Z}_{\geq 0}$. Completed simulation runs in a payment sequence; each run contains T periods.
I_q, J_q	$\{0, 1\}$. Income and rent payment indicators at run boundary q, defined in Section 6.1.
p_0	$\mathbb{R}_{>0}$. Reference home price.
σ, ℓ	$\mathbb{R}_{\geq 0}$. Log-price variation and spatial smoothing scale.
ρ	$\mathbb{Z}_{>0}$. Mean homes per region.
σ_h, W_ω	$\mathbb{R}_{\geq 0}$ and $\mathbb{R}$. Log-price spread among homes in a region, and home ω's price noise.
$\zeta, \ell_{\text{size}}$	$\mathbb{R}_{\geq 0}$. Region-size contrast and spatial smoothing scale.
H_{ab}	$\mathbb{Z}_{>0}$. Homes in region (a, b), defined in Section 5.
Y_e, r_e	$\mathbb{R}_{\geq 0}$ and $\mathbb{Z}_{\geq 0}$. Tweedie draw and radius of disaster e, defined in Section 9.2.
μ_r, ϕ	$\mathbb{R}_{>0}$. Mean and dispersion of the Tweedie draw, set to 0.3 and 1.5.
τ	$\mathbb{R}_{\geq 0}$. Noise strength for residence placement.
λ_t	$\mathbb{R}_{\geq 0}$. Expected disasters in period t.
S	$\mathbb{Z}_{\geq 0}$. Global seed for labeled component random streams.

Agents are indexed by nondecreasing income, with seeded tie-breaking.

4 Assigning initial home values

Prices vary by region, and every home in a region starts at that region's price. Let $E \in \mathbb{R}^{m \times n}$ be the seeded noise matrix used to generate them:

$$E_{ab} \overset{\text{iid}}{\sim} \mathcal{N}(0, 1). \tag{3}$$

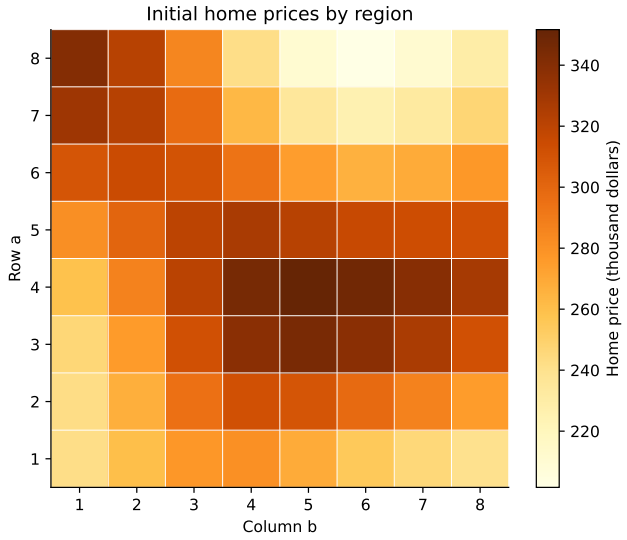


Figure 3: Each cell is one region; color shows the mean initial price of its homes, generated from seeded spatial noise. Example: $m = n = 8$, $S = 42$, $p_0 = 250{,}000$, $\sigma = 0.4$, $\ell = 2$.

We reuse this smoothing operator for disaster risk in Section 9.1. For scale $\ell > 0$, let $w_{ab,uv}$ weight cell (u, v) at location (a, b). Define $\mathcal{S}_\ell$ by

$$w_{ab,uv} = \exp\left(-\frac{(a-u)^2 + (b-v)^2}{2\ell^2}\right),$$

$$[\mathcal{S}_\ell(E)]_{ab} = \frac{\sum_{u=1}^{m}\sum_{v=1}^{n} w_{ab,uv} E_{uv}}{\sqrt{\sum_{u=1}^{m}\sum_{v=1}^{n} w_{ab,uv}^2}}. \tag{4}$$

The scale ℓ controls neighborhood size; normalization gives unit variance at each cell. Set $\mathcal{S}_0(E) = E$ for no smoothing.

Using reference price p_0 and log-price spread σ, the regional price matrix $\bar{P} \in \mathbb{R}_{>0}^{m\times n}$ and initial home prices are

$$\begin{aligned} \bar{P} &= p_0 \exp\left(\sigma \mathcal{S}_\ell(E)\right), \\ P_\omega &= \bar{P}_{a_\omega b_\omega} \frac{H_{a_\omega b_\omega} \exp(\sigma_h W_\omega)}{\sum_{\omega' : (a_{\omega'}, b_{\omega'}) = (a_\omega, b_\omega)} \exp(\sigma_h W_{\omega'})}, \quad W_\omega \overset{\text{iid}}{\sim} \mathcal{N}(0,1). \end{aligned} \tag{5}$$

Algorithm 1 Generate initial home values

Input: Grid dimensions, region sizes, p_0, σ, ℓ, and seeded price stream

1: Draw $E_{ab} \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$
2: $\bar{P} \leftarrow p_0 \exp\big(\sigma \mathcal{S}_\ell(E)\big)$; draw W_ω and set P_ω by the equation above; $D \leftarrow 0$
3: **return** P, D

The entrywise exponential gives positive regional prices with median p_0 at each cell. Homes in one region have different prices, and this is an assumption: $\sigma_h \geq 0$ sets their log-price spread, 0.2 by default, and the normalization makes each region's mean home price equal $\bar{P}_{ab}$ exactly, before rounding. A region with one home has price $\bar{P}_{ab}$. Prices are independent of population allocation and rounded to cents.

5 Assigning homes to regions

Earlier versions placed one home, and so one agent, in every grid cell. We now treat each cell as a region whose size is its number of homes. This is a modeling assumption: size and home count are the same quantity here, so land area is not modeled separately. The command center therefore draws the map as a treemap: each region's area is proportional to its home count, regions keep their relative grid positions, and each region shows the mean of the selected quantity over its homes.

Let $G \in \mathbb{R}^{m\times n}$ be seeded noise independent of price and risk noise, $\zeta \geq 0$ the region-size contrast, and $\ell_{\text{size}} \geq 0$

its smoothing scale. Reusing $\mathcal{S}$ from Section 4, define the weights A and the extra homes X beyond each region's first:

$$
\begin{aligned}
G_{ab} &\overset{\text{iid}}{\sim} \mathcal{N}(0,1), && \text{size noise,} \\
A_{ab} &= \exp(\zeta[\mathcal{S}_{\ell_{\text{size}}}(G)]_{ab}), && \text{size weight,} \\
X_{ab} &\approx (H - mn)\,\frac{A_{ab}}{\sum_{u=1}^{m}\sum_{v=1}^{n} A_{uv}}, && \text{extra homes,} \\
H_{ab} &= 1 + X_{ab}, && \text{homes in the region.}
\end{aligned}
\tag{6}
$$

The whole-number counts X_{ab} use the rounding and residual correction of Algorithm 2, so $\sum_{a,b} H_{ab} = H$ exactly and every region keeps at least one home. Smoothing keeps neighboring regions similar in size, so dense clusters form. At $\zeta = 0$ every region holds ρ homes, and at $\rho = 1$ every region holds one home, which recovers the earlier grid.

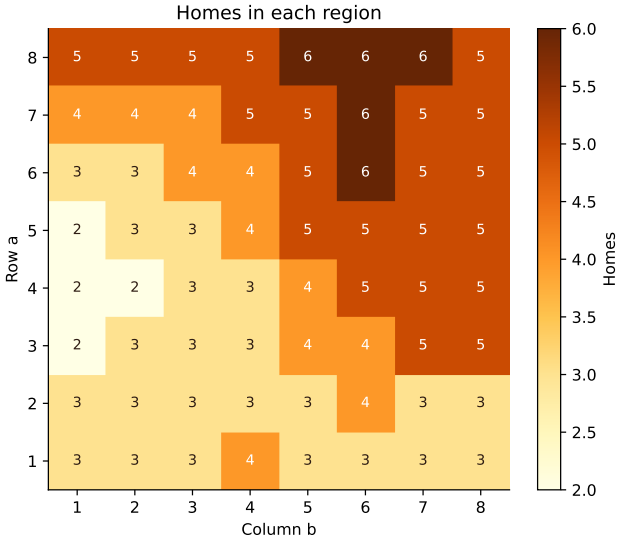


Figure 4: Illustrative region sizes. Example: $m = n = 8$, $\rho = 4$, $H = 256$, $\zeta = 1$, $\ell_{\text{size}} = 2$.

Homes have no positions inside a region, so a disaster strikes all of a region's homes or none of them. All three worlds share the same region sizes.

6 Assigning income and ownership shares

The Lorenz curves L_y and L_h allocate $N\mu$ annual income and H homes along a shared income ranking. Separate curves control income and ownership concentration.

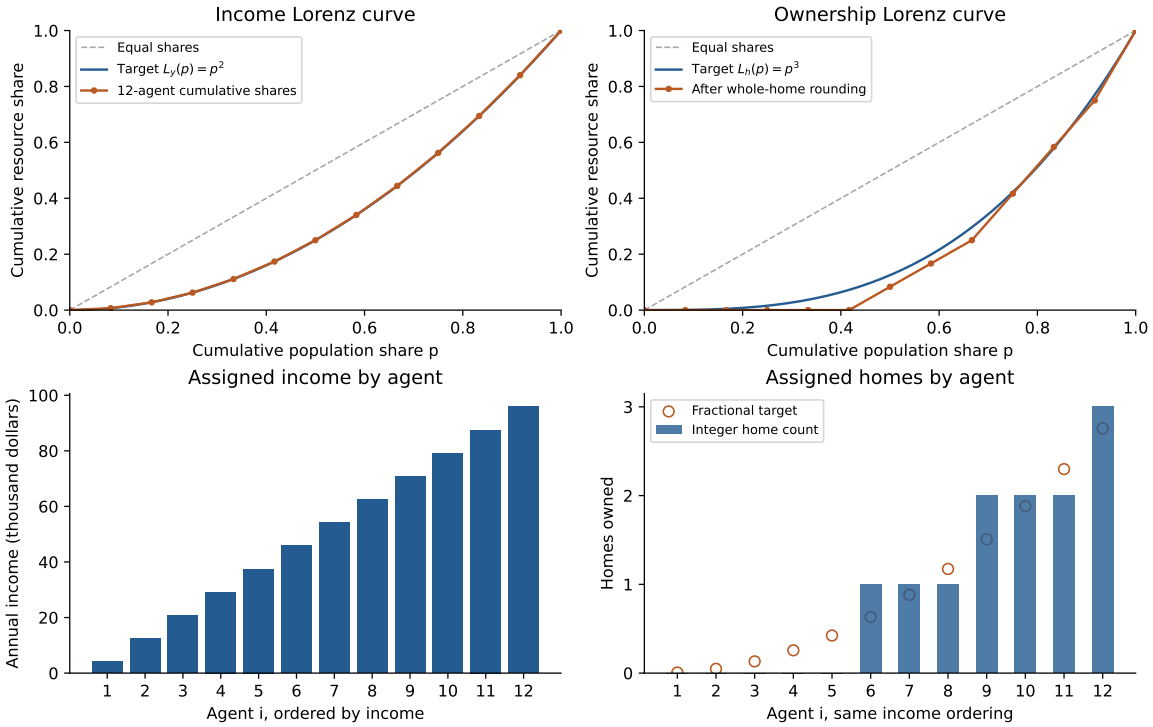


Figure 5: Illustrative allocation: $N = H = 12$, $\mu = 50{,}000$, $L_y(p) = p^2$, $L_h(p) = p^3$. Markers show finite-population shares; lower panels show incomes and integer home counts against fractional targets.

For $L \in \{L_y, L_h\}$, let $\mathbf{d}(L) \in [0,1]^N$ contain individual shares:

$$[\mathbf{d}(L)]_i = L\left(\frac{i}{N}\right) - L\left(\frac{i-1}{N}\right), \qquad i \in \{1, \ldots, N\}. \tag{7}$$

Let y_i be annual income and a_i the fractional home target for agent i:

$$\begin{aligned} y_i &= N\mu\left[L_y\left(\frac{i}{N}\right) - L_y\left(\frac{i-1}{N}\right)\right], \\ a_i &= H\left[L_h\left(\frac{i}{N}\right) - L_h\left(\frac{i-1}{N}\right)\right]. \end{aligned} \tag{8}$$

Let $\mathbf{a} \in \mathbb{R}^N_{\geq 0}$ collect fractional targets and $\mathbf{h} \in \mathbb{Z}^N_{\geq 0}$ whole-home allocations. Round and correct the total:

1. Round each a_i to the nearest integer, rounding halves upward.
2. If $\sum_i h_i < H$, add one to the count with largest $a_i - h_i$.
3. If $\sum_i h_i > H$, subtract one from a positive count with largest $h_i - a_i$.

Algorithm 2 Assign incomes and home counts

Input: N, H, μ, L_y, L_h and seeded tie-breaking

$\mathbf{y} \leftarrow N\mu\mathbf{d}(L_y)$; $\mathbf{a} \leftarrow H\mathbf{d}(L_h)$
Round each a_i to the nearest integer h_i, with halves rounded upward
Correct largest rounding residuals until $\sum_i h_i = H$, as specified above
return $\mathbf{y}, \mathbf{h}$

Repeat until $\sum_i h_i = H$, recomputing residuals and breaking ties by seed. Ownership shares target home counts, not property values.

6.1 Payment schedule across simulation runs

Let q count completed simulation runs (episodes), each lasting $T\Delta t$ years. Income is paid every 10 runs and rent every 20:

$$\begin{aligned} I_q &= \mathbf{1}_{\{q>0 \text{ and } q \bmod 10 = 0\}}, \quad \text{income payment,} \\ J_q &= \mathbf{1}_{\{q>0 \text{ and } q \bmod 20 = 0\}}, \quad \text{rent payment.} \end{aligned} \tag{9}$$

Income accumulates between paydays:

$$\mathbf{c} \leftarrow \mathbf{c} + I_q \, (10T\Delta t) \, \mathbf{y}. \tag{10}$$

When $J_q = 1$, tenants transfer accrued rent to their landlords, after income is credited. Cash and unpaid balances carry between runs; independent sequences start at $q = 0$ with no payment due. Rent amounts and insufficient-cash rules remain open.

7 Connecting agents to homes

Residences are assigned by income with random variation; ownership is assigned separately.

For agent i, let B_i contain agents with lower income and T_i contain agents with the same income:

$$\begin{aligned} B_i &:= \{j \in \{1, \ldots, N\} : y_j < y_i\}, && \text{lower-income agents,} \\ T_i &:= \{j \in \{1, \ldots, N\} : y_j = y_i\}, && \text{equal-income agents.} \end{aligned}$$

Let $r_i \in [0, 1]$ be the income percentile, $z_i \in \mathbb{R}$ an independent noise draw, and $q_i \in \mathbb{R}$ the placement score. Using $|B_i|$ and $|T_i|$ for the respective group sizes, define

$$\begin{aligned} r_i &:= \frac{|B_i| + \frac{1}{2}|T_i|}{N}, && \text{income percentile,} \\ z_i &\overset{\text{iid}}{\sim} \mathcal{N}(0, 1), && \text{placement noise,} \\ q_i &:= r_i + \tau z_i, && \text{placement score.} \end{aligned} \tag{11}$$

Match agents ranked by $\mathbf{q}$ to homes ranked by price. Equal incomes receive equal percentiles; ties are broken by seed. Increasing τ weakens income-based sorting. An independent baseline uses random assignment.

Each agent with $h_i > 0$ first owns their residence. Shuffle the remaining owner-capacity slots across unassigned titles. Larger residual quotas receive proportionally more titles; agents with $h_i = 0$ become renters.

Algorithm 3 Assign residences and ownership

Input: $P, \mathbf{y}, \mathbf{h}, \tau$ and seeded placement and ownership streams

1: Compute income percentiles $\mathbf{r}$ and draw $z_i \overset{\text{iid}}{\sim} \mathcal{N}(0, 1)$
2: Match agents ranked by $q_i = r_i + \tau z_i$ to homes ranked by price, filling R
3: Give each agent with $h_i > 0$ ownership of their residence
4: Shuffle residual owner-capacity slots and assign remaining titles, filling O
5: **return** R, O

8 Insurance coverage and claims

Insurers offer homeowners, landlord, and renters coverage. The opening round collects offers and coverage choices before exposure. Fixed quotes and ownership–price portfolio assignments provide the current baseline for learned pricing and purchasing.

For a covered claim, let $x \geq 0$ be the loss amount, $\delta \geq 0$ the deductible, and $L_{\max} \geq 0$ the coverage limit. Let $b_f \geq 0$ be insurer f's available capital immediately before payment, with $f \in \{1, \ldots, F\}$. All amounts are in dollars.

Let C_{due} denote the contractual payment, C_{paid} the actual payment, and C_{short} the unpaid contractual amount:

$$\begin{aligned} C_{\text{due}} &= \min\{L_{\max}, \max\{0, x - \delta\}\}, && \text{payment due,} \\ C_{\text{paid}} &= \min\{C_{\text{due}}, b_f\}, && \text{payment made,} \\ C_{\text{short}} &= C_{\text{due}} - C_{\text{paid}}, && \text{capital shortfall.} \end{aligned} \tag{12}$$

For a claim on home ω, let P_ω^{pre} be its value immediately before destruction and $v_{\text{contents}} \geq 0$ the contract's contents-loss amount. Under the total-loss rule,

$$x = \begin{cases} P_\omega^{\text{pre}}, & \text{homeowners or landlord building claim,} \\ v_{\text{contents}}, & \text{renters contents claim.} \end{cases} \tag{13}$$

For insured agent i, settlement transfers the payment:

$$\begin{aligned} b_f &\leftarrow b_f - C_{\text{paid}}, && \text{insurer capital,} \\ c_i &\leftarrow c_i + C_{\text{paid}}, && \text{insured agent cash.} \end{aligned} \tag{14}$$

Claims are settled in contract-ID order using the remaining capital for each payment.

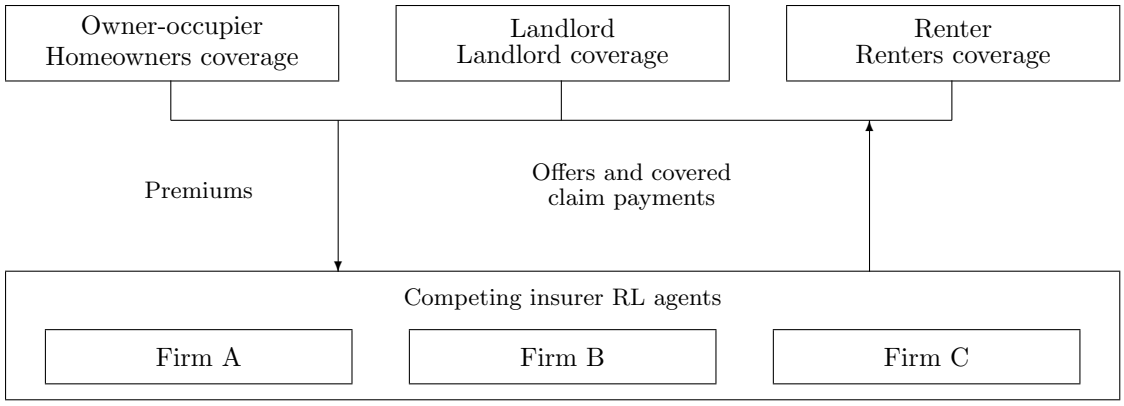


Figure 6: Housing roles determine applicable coverage. Agents choose among competing firms; arrows represent market access and contractual payments.

Owners and tenants may insure different losses at the same home with different firms.

Algorithm 4 Open the insurance market

Input: Housing allocation, cash, insurer capital, and decision policies
1: Collect all insurer offers, then each agent's coverage choices
2: Validate eligibility and affordability; transfer accepted premiums
3: Activate contracts and return updated balances

8.1 Implemented initialization: ownership–price insurer portfolios

The command center identifies insurers as Red (A), Green (B), and Blue (C). All start with equal capital and retain fixed premiums of 125, 150, and 175 respectively. Opening purchases now use the home's assigned insurer rather than the globally cheapest firm. This initialization is implemented in all three experimental worlds and their regenerated snapshots.

The assignment reuses actual ownership and initial prices; it does not draw a new spatial field or move home titles. For each owner i with at least one home, compute the initial portfolio value

$$W_i = \sum_{\omega : O_\omega = i} P_\omega(0). \tag{15}$$

Rank these owners by W_i, breaking ties with a dedicated seeded insurance stream. Let M be the number of such owners. Use the existing ownership Lorenz curve L_h, evaluated over the F firms, to obtain

$$w_j = L_h(j/F) - L_h((j-1)/F), \qquad j = 1, \ldots, F. \tag{16}$$

When $M \geq F$, reserve one owner portfolio per firm and allocate the remaining $M - F$ portfolios using w_j and the existing count-rounding and residual-correction procedure. Thus each firm has at least one owner portfolio, and counts sum exactly to M. When $M < F$, allocate all M portfolios using the same rounding rule without a reservation; some firms necessarily receive none. These are counts of whole owner portfolios, not target home counts, policy counts, or insured dollar shares. The minimum-one reservation intentionally modifies the raw Lorenz shares.

A seeded permutation assigns stable firm identities to the ordered quotas. Assign consecutive blocks of value-ranked owners to those firms, then define the field

$$I_\omega = \text{assigned insurer of owner } O_\omega. \tag{17}$$

Every home owned by the same person therefore has the same assigned insurer, including geographically separated rental properties. Higher-ranked blocks contain higher-value ownership portfolios; firm color is not an intrinsic wealth rank. In "Peace", $L_h(p) = p$ and equal values use seeded ties, giving nearly equal portfolio counts. "Standard" and "Chaos" share the same ownership, prices, seed, and insurer field. The field is independent of disaster sampling and stays fixed after damage.

Owners and renters purchase their eligible products from the assigned insurer if they can afford the premium. Building and contents coverage remain distinct contracts at the same home. Purchases proceed in home-ID order within each person's cash budget. An unaffordable assigned offer results in no contract, with no fallback to a cheaper competing firm. Firms receive premiums and pay capital-limited claims through the existing insurance

settlement rules. Assignment does not guarantee active coverage or equal exposure, and insurers do not become owners of the homes.

Snapshots record the home-to-insurer mapping and initialization version. The command center's insurer-portfolio layer displays the initial assignment separately from active coverage; firm controls show current policy counts. The dropdown separates ownership and resident views: in ownership view, fills and IDs identify owners, including rented homes; in resident view, fills and IDs identify current occupants. Each person has a stable unique hex color across both views. Home borders are neutral by default. Selecting an insurer highlights its assigned homes with that firm's Red, Green, or Blue border while all other homes remain fully visible; only one insurer is selected at a time. Clearing the selection restores neutral borders. Portfolio highlights persist after damage or coverage expiry and do not assert an active contract; the Coverage view highlights only current coverage. Existing ownership and price maps, initial capital budgets, and disaster streams are preserved. The previous cheapest-affordable policy remains available as a separate baseline, but is no longer the initialization used by these three worlds.

9 Disaster occurrence and damage

Each disaster affects a square zone of regions, with a random center and a random size. All intact homes in those regions lose all modeled value and become uninhabitable; homes outside it are unaffected.

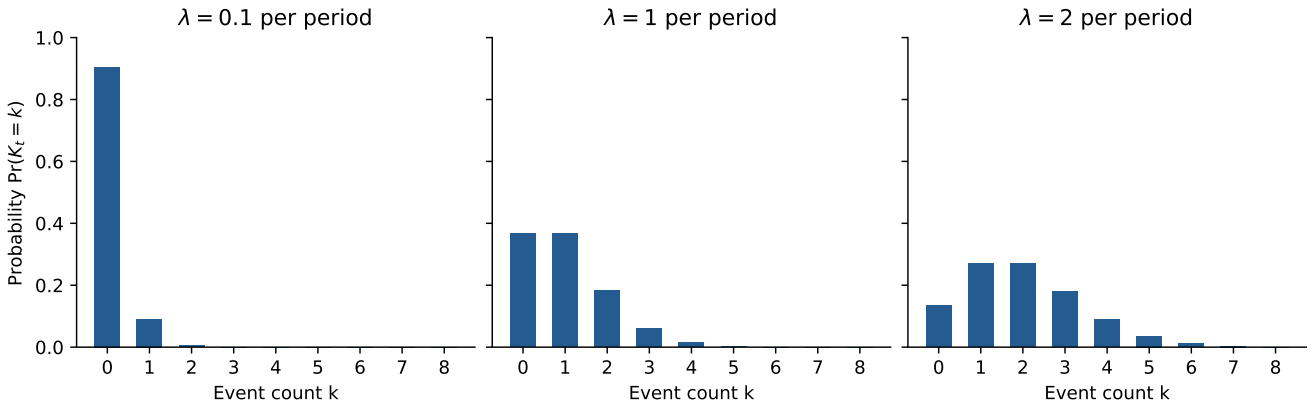


Figure 7: Disaster counts per period for illustrative rates, not selected simulation parameters. Displayed probability masses cover $k \in \{0, \ldots, 8\}$; the omitted tail is below 0.00024 in each panel.

Let $K_t \in \mathbb{Z}_{\geq 0}$ count disasters in period t, with mean λ_t, and let $f(k; \lambda_t)$ be its probability mass function. Counts are independent conditional on the rate schedule:

$$\begin{aligned} K_t &\sim \text{Poisson}(\lambda_t), \\ f(k; \lambda_t) = \Pr(K_t = k) &= \frac{\lambda_t^k e^{-\lambda_t}}{k!}, \qquad k \in \mathbb{Z}_{\geq 0}. \end{aligned} \tag{18}$$

At $\lambda_t = 0$, $\Pr(K_t = 0) = 1$.

9.1 Hidden spatial risk

A center is the region in the middle of a disaster zone. Every region can be a center.

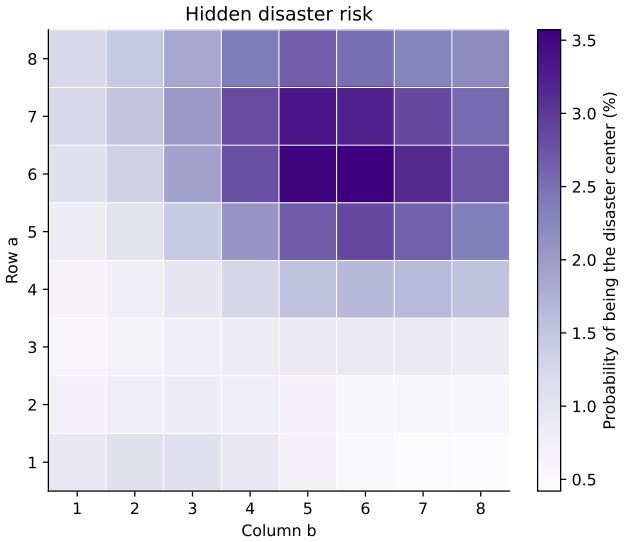


Figure 8: Darker purple denotes a higher probability of being the disaster center. Example: 8×8 regions, $S = 42$, $\eta = 1$, $\ell_{\text{risk}} = 2$. This risk map is hidden from agents.

Let $U \in \mathbb{R}^{m \times n}$ be seeded Gaussian noise independent of price noise, $\ell_{\text{risk}} \geq 0$ the smoothing scale, and $\eta \geq 0$ the risk contrast. Reusing $\mathcal{S}$ from Section 4, define the positive weight field V:

$$\begin{aligned} U_{ab} &\overset{\text{iid}}{\sim} \mathcal{N}(0,1), && \text{risk noise,} \\ V_{ab} &= \exp(\eta[\mathcal{S}_{\ell_{\text{risk}}}(U)]_{ab}), && \text{spatial weight.} \end{aligned} \tag{19}$$

Let π_{ab} be the probability of selecting center (a, b), and $\lambda_t(a, b)$ its expected event count in period t:

$$\begin{aligned} \pi_{ab} &= \frac{V_{ab}}{\sum_{u=1}^{m}\sum_{v=1}^{n} V_{uv}}, && \text{location probability,} \\ \lambda_t(a,b) &= \lambda_t \pi_{ab}, && \text{time–space event intensity.} \end{aligned} \tag{20}$$

Normalization preserves the total rate λ_t. Draw the Poisson count, then sample centers independently from π. At $\eta = 0$, centers are equally likely; larger η concentrates events. Zones reach past their centers, so a region's exposure, given in Section 9.2, is larger than its own π_{ab}.

The field stays fixed throughout an episode. All agents observe event times, zones, and damage, including events on previously destroyed homes; U, V, and π remain hidden.

9.2 Disaster size

Disaster sizes follow a Tweedie distribution. We previously used a standard normal, but it made zones too large: 31.7% of events reached 3×3 or wider, and an average event struck more than four regions. A Tweedie with power between 1 and 2 has a point mass at zero, so many disasters have no spread at all, and its dispersion directly limits how often large zones occur. This is an assumption. We fix the power at 3/2 because the sum below then contains only exponential variables, which keeps the math simple. For event e with center (u, v), let Y_e be its draw, r_e its radius, and $\mathcal{F}_e$ its zone:

$$\begin{aligned} M_e &\overset{\text{iid}}{\sim} \text{Poisson}(\psi), \quad \psi = \frac{2\sqrt{\mu_r}}{\phi}, && \text{number of jumps,} \\ Y_e &= \sum_{j=1}^{M_e} J_{ej}, \quad J_{ej} \overset{\text{iid}}{\sim} \text{Exponential(mean } \nu), \quad \nu = \frac{\phi\sqrt{\mu_r}}{2}, && \text{size draw,} \\ r_e &= \lfloor Y_e \rfloor, && \text{radius in regions,} \\ \mathcal{F}_e &= \{(a,b) : \max\{|a-u|, |b-v|\} \leq r_e\}, && \text{affected regions.} \end{aligned} \tag{21}$$

Then $\mathbb{E}[Y_e] = \mu_r$ and $\text{Var}(Y_e) = \phi\mu_r^{3/2}$, the defining mean–variance relation of a Tweedie distribution with power 3/2. Zones have side length $2r_e + 1$ and are clipped at the grid edge. Sizes are independent of counts, centers, and

the risk field. Lowering ϕ makes large zones rarer, and both μ_r and ϕ are settings of each experiment.

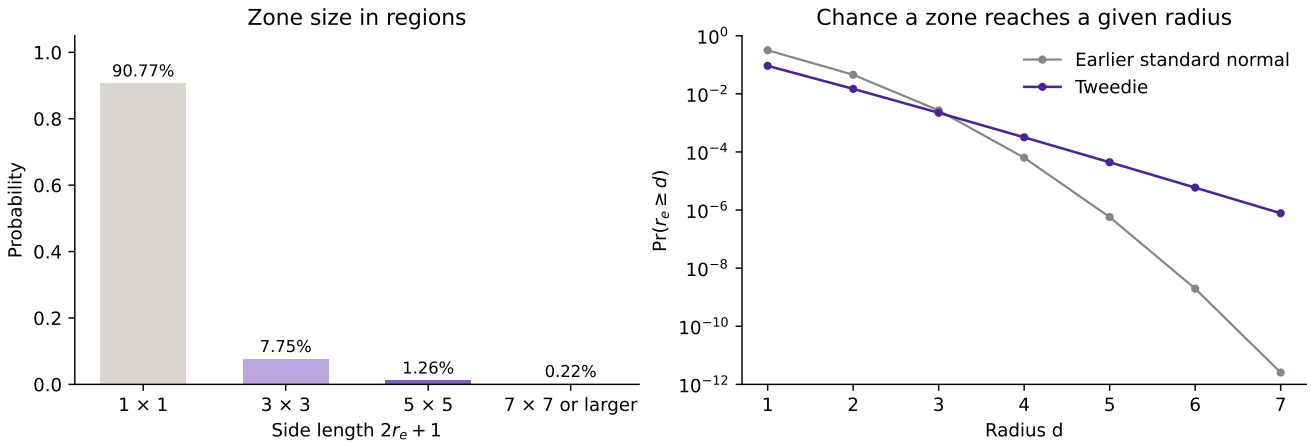


Figure 9: Disaster sizes with $\mu_r = 0.3$ and $\phi = 1.5$. Left: probability of each zone size. Right: the chance a zone reaches radius d, on a logarithmic scale, compared with the earlier standard normal.

With these values, 90.8% of events strike a single region, an average event strikes 2.04 regions before clipping, and a zone of 7×7 or larger occurs in about 1 of every 448 events. Beyond radius 3 the Tweedie tail decays exponentially, more slowly than the normal's, so it keeps rare outliers while the common sizes shrink.

Because $\lfloor Y \rfloor \geq d$ exactly when $Y \geq d$, a region's exposure can be computed exactly. A sum of k exponential variables exceeds d exactly when fewer than k arrivals of a Poisson process with rate $1/\nu$ occur by time d, so for $d \geq 1$

$$\xi(d) := \Pr(r_e \geq d) = \sum_{k=1}^{\infty} \frac{e^{-\psi}\psi^k}{k!} \sum_{i=0}^{k-1} \frac{e^{-d/\nu}(d/\nu)^i}{i!}, \qquad \xi(0) = 1. \tag{22}$$

Let $d_{ab,uv} = \max\{|a-u|, |b-v|\}$. The probability that one event strikes region (a, b), and its expected strikes in period t, are

$$\theta_{ab} = \sum_{u=1}^{m} \sum_{v=1}^{n} \pi_{uv}\, \xi(d_{ab,uv}),$$
$$\Lambda_t(a, b) = \lambda_t \theta_{ab}. \tag{23}$$

Strikes on a region are Poisson with mean $\Lambda_t(a, b)$, so it is struck at least once in period t with probability $1 - \exp(-\Lambda_t(a, b))$. Every home in a region shares this hazard. Larger regions are no more likely to be struck, but more homes and claims are at stake when they are.

9.3 Damage and claims

For event $e \in \{1, \ldots, K_t\}$, consider each home ω with $(a_\omega, b_\omega) \in \mathcal{F}_e$ and $D_\omega = 0$. Retain its pre-loss records and set

$$P_\omega \leftarrow 0,$$
$$R_\omega \leftarrow 0, \tag{24}$$
$$D_\omega \leftarrow 1.$$

This update displaces the resident and removes the owner's property value while retaining title. Pre-loss values and occupancy determine covered losses. One zone can trigger claims across several owners, tenants, and insurers.

Algorithm 5 Apply disasters and claims

Input: P, O, R, D, contracts, λ_t, π, and disaster streams

Draw $K_t \sim \text{Poisson}(\lambda_t)$
for $e \in \{1, \ldots, K_t\}$ **do**
    Sample a center from π and draw Y_e from the Tweedie distribution
    Form zone $\mathcal{F}_e$ with radius $r_e = \lfloor Y_e \rfloor$, clipped at the grid edge
    Retain pre-loss records for homes in $\mathcal{F}_e$ where $D_\omega = 0$
    For those homes: $P_\omega \leftarrow 0$, $R_\omega \leftarrow 0$, $D_\omega \leftarrow 1$
    Record displacement and settle covered claims; retain ownership

8: Publish event time, zone, and housing damage to all agents
9: **end for**

The environment above supplies the housing, insurance, and disaster outcomes used by the proposed learning agents. The following section specifies their states, actions, and rewards; Section 11 then defines the experimental worlds in which these learners will be compared.

10 Tabular Q-learning for the two agent types

People and insurers choose from different actions, so we use two tabular Q-learners, Q^h and Q^f, with separate states, action sets $\mathcal{A}^h, \mathcal{A}^f$, discount factor γ, and rewards r. For type $k \in \{h, f\}$, both use

$$Q^k(s,a) \leftarrow Q^k(s,a) + \alpha\Big[r + \gamma \max_{a'} Q^k(s',a') - Q^k(s,a)\Big], \tag{25}$$

with ε-greedy action selection and one shared table per type. All specifications below are proposals.

Households. The state is $s_i^h = (\text{role}_i, \text{current insurer}_i, \text{displaced}_i, \text{house price/rent price}_i)$, where the role is owner-occupier, renter, landlord, or none, and the insurer is one of $1, \ldots, F$ or none. A person takes one action per period. Opting in selects one insurer's offer for the product their status in O and R implies, with the premium deducted from income each period; only one policy is active at a time.

$$\begin{aligned}\mathcal{A}^h = \{&\text{buy home, sell home, pay rent, rent out home, hold,}\\ &\text{opt in to insurer 1, } \ldots, \text{ opt in to insurer } F, \text{ opt out}\}.\end{aligned} \tag{26}$$

Actions are limited by role and cash, and buying, selling, and rent still require transaction rules, which remain open. The reward is the change in net position, after premiums, less a displacement penalty $\kappa \geq 0$:

$$r_i^h = \Delta c_i + \Delta\big(\text{property wealth}_i\big) - \kappa\, \mathbf{1}_{\{\text{displaced}_i\}}. \tag{27}$$

Insurers. With limits and deductibles fixed, a firm adjusts one product's premium per period on a bounded ladder of multiples of a fitted pure premium, so $\mathcal{A}^f = \{\text{raise, lower, keep}\}$. The pure premium comes from a Tweedie GLM fit offline to regional loss data, which suits losses that are zero in most periods and skewed otherwise. Idle capital earns a risk-free rate $r_f \geq 0$, so $\text{capital}_j \leftarrow (1 + r_f)\,\text{capital}_j + \text{premiums}_j - \text{claims}_j$. The state and reward are

$$s_j^f = \big(\text{premium level}_j,\ \mathbf{1}_{\{\text{capital}_j > 0\}}\big), \qquad r_j^f = \text{premiums}_j - \text{claims}_j - \kappa_f\, \mathbf{1}_{\{\text{capital}_j \leq 0\}}, \tag{28}$$

with insolvency penalty $\kappa_f \geq 0$. Competitors' premiums change what households buy, so each firm faces a non-stationary environment, which we treat as part of the study.

11 Environment Variants

The worlds "Peace", "Standard", and "Chaos" compare initial resource inequality and disaster-rate schedules.

11.1 Shared experimental controls

Fix the population N, insurer count F, housing supply $H = N$, episode length $T \in \mathbb{Z}_{>0}$, and period duration Δt. There are $N + F$ RL agents. Let $M_0 \in \mathbb{R}_{\geq 0}$ be the total initial cash across agents and insurers, and let $b_f(0)$ denote insurer f's initial capital. Let $Y \in \mathbb{R}_{>0}$ be the population's total annual external income. These budgets satisfy

$$\begin{aligned} M_0 &= \sum_{i=1}^{N} c_i(0) + \sum_{f=1}^{F} b_f(0), && \text{initial cash budget,}\\ Y &= \sum_{i=1}^{N} y_i = N\mu, && \text{annual income budget.}\end{aligned} \tag{29}$$

M_0 fixes initial cash; recurring income increases it. Y is an annual income budget, not a full GDP measure.

Matched runs share budgets, insurer capital, region sizes, the disaster-size distribution, risk field, insurance rules, learning configuration, and seeds. All three worlds use the same 10-run income and 20-run rent schedule in Section 6.1; comparisons use equal numbers of runs and the same payment-clock phase. Initial cash budgets apply at the start of each independent payment sequence. Vary η and ℓ_{risk} separately to test spatial concentration.

11.2 "Peace": equal initial resources

"Peace" starts every agent with the same income, one home, and the same cash balance. Every home has the same initial price, so $\sigma = \sigma_h = 0$. Let $M_{\text{agents}} := M_0 - \sum_{f=1}^{F} b_f(0)$ be the nonnegative cash budget assigned to agents. For population share $p \in [0,1]$, choose

$$\begin{aligned} L_y(p) = L_h(p) = p, &\qquad && \text{equal resource shares,} \\ y_i = \frac{Y}{N}, \qquad h_i = 1, &\qquad && \text{income and home allocation,} \\ c_i(0) = \frac{M_{\text{agents}}}{N}, &\qquad && \text{equal initial cash,} \\ P_\omega(0) = p_0, \qquad \sigma = 0, &\qquad && \text{equal home values.} \end{aligned} \tag{30}$$

Everyone initially owns their residence and has equal wealth. Subsequent decisions and losses may produce inequality; external income remains fixed. There is no wealth cap or redistribution.

"Peace" shares the constant rate, region sizes, and hidden risk field of "Standard". Initial resources are equal; geographic exposure varies.

11.3 "Standard": unequal resources and constant risk

"Standard" uses the Lorenz allocations, spatial prices, and residence assignment above. Tune $L_y, L_h, \sigma, \ell, \tau$ while preserving total income and home count. Initial cash remains equal unless tested separately.

Let $\bar{\lambda} \in \mathbb{R}_{>0}$ be the selected constant expected event count per period. The reference risk schedule is

$$\lambda_t = \bar{\lambda}, \qquad t \in \{0, \dots, T-1\}. \tag{31}$$

Select $\bar{\lambda}$ relative to episode length and the mean disaster size; numerical calibration remains open.

11.4 "Chaos": exponentially increasing risk

"Chaos" shares the initial allocation and spatial probabilities of "Standard". Let $\lambda_0 > 0$ be its low initial event rate and $g > 0$ the growth per step:

$$\begin{aligned} \lambda_t = \lambda_0 \exp(gt), &\qquad t \in \{0, \dots, T-1\}, \\ K_t \sim \text{Poisson}(\lambda_t), &\quad \text{events in period } t. \end{aligned} \tag{32}$$

The rate grows by $\exp(g)$ per step, producing larger absolute increases later. Realized counts remain Poisson and need not increase every period.

To tune the onset, choose a target rate $\lambda_\star > \lambda_0$ at a step $t_\star \in \{1, \dots, T-1\}$, for $T \geq 2$. The required growth parameter is

$$g = \frac{\log(\lambda_\star / \lambda_0)}{t_\star}. \tag{33}$$

Without rebuilding, later events may revisit destroyed homes. Track new losses and displacement as well as event counts.

11.5 Research comparisons

Compare "Peace" with "Standard" for initial inequality, and "Standard" with "Chaos" for increasing risk. To isolate timing, match the constant rate in "Standard" to the mean rate in "Chaos":

$$\bar{\lambda} = \frac{1}{T} \sum_{t=0}^{T-1} \lambda_0 \exp(gt). \tag{34}$$

This matches expected event totals. Across seeds, compare coverage choices, holdings, wealth concentration, displacement, and insurer solvency. Vary individual parameters for ablations.

12 Questions for evaluation

How does initial inequality affect housing and coverage decisions? How do agents learn regional risk from public disasters, and how does that learning affect insurance pricing and losses?